SELENIUM & TestNG EXAM GUIDE

Complete Study Notes - Simple & Easy Format

PART 1: SELENIUM

1. What is Selenium?

Selenium is a free, open-source automation tool used to test web applications. It helps testers automate boring repetitive tasks and test websites across different browsers and operating systems.

2. Advantages, Disadvantages, Characteristics & Goals

✓ ADVANTAGES:

- Free and open-source (no cost)
- Works on multiple browsers (Chrome, Firefox, Safari, Edge)
- Supports multiple programming languages (Java, Python, C#, Ruby, etc.)
- Works on different operating systems (Windows, Mac, Linux)
- Can test complex web applications
- Large community support

✗ DISADVANTAGES:

- Cannot test desktop applications
- Cannot test mobile applications (limited support)
- Cannot do API testing
- Cannot handle pop-ups (alerts can be handled but not all pop-ups)
- Requires good coding knowledge

⭐ CHARACTERISTICS:

- Open-source
- Supports cross-browser testing
- Lightweight and fast
- User-friendly

🎯 GOALS:

- Reduce testing time
- Increase test coverage
- Reduce manual testing effort
- Improve software quality

3. Selenium Components

Selenium has 4 main components:

- **Selenium IDE (Integrated Development Environment):**
  A Firefox plugin for record and playback. Records user actions and creates test scripts automatically.
- **Selenium WebDriver (RC - Remote Control):**
  Uses browser drivers to directly communicate with browsers. Most popular component. Used for writing automated test scripts.

- **Selenium Grid:**
  Allows tests to run on multiple machines and browsers in parallel (at same time).
- **Selenium Remote Control (Deprecated):**

  Old component. Now replaced by WebDriver.

4. Locators in Selenium (8 Locators)

Locators are used to find web elements on a page. Here are the 8 main locators:

Locator Type	How to Use
ID	Find element by its unique ID attribute
Name	Find element by its name attribute
Class Name	Find element by its CSS class
CSS Selector	Find element using CSS syntax (most powerful)
XPath	Find element using XML path (most flexible)
Link Text	Find link element by exact text
Partial Link Text	Find link element by partial text match
Tag Name	Find element by HTML tag name

5. Types of Testing in Selenium

- **Functional Testing:**
  Test if application features work correctly
- **Smoke Testing:**
  Test basic functionality to ensure nothing is broken
- **Regression Testing:**
  Test again to make sure new changes don't break old functionality
- **Sanity Testing:**
  Quick test to check if a specific bug fix works
- **Cross-browser Testing:**
  Test application on different browsers

6. Architecture of Selenium

Selenium has 4 main layers:

1. **Language Bindings:**
   Java, Python, C#, Ruby, JavaScript, etc.
2. **Selenium Client Library:**
   APIs for communicating with browsers
3. **Browser Drivers:**

ChromeDriver, FirefoxDriver, EdgeDriver, SafariDriver

4. **Browsers:**

   Chrome, Firefox, Safari, Edge

7. DOM & POM in Selenium

DOM (Document Object Model):

DOM is a tree structure that represents all HTML elements on a web page. Selenium uses DOM to find and interact with web elements.

POM (Page Object Model):

POM is a design pattern for organizing test code. Each web page has a class with all its elements and methods.

Advantages of POM:

- Easy to maintain
- Improves test readability
- Reduces code duplication
- Makes tests more reliable

8. Wait Types in Selenium

Sometimes web elements take time to load. Waits tell Selenium to wait before finding elements.

1️⃣ **IMPLICIT WAIT (Global Wait):**

- Applies to all elements
- Waits for a specific time (e.g., 10 seconds)
- Sets once and works for entire script
- Slower - waits full time even if element is found quickly

2️⃣ **EXPLICIT WAIT (Specific Wait):**

- Applied to specific elements only
- Waits for a condition to be true
- Can wait for element to be clickable, visible, etc.
- Faster - moves forward as soon as condition is met

9. Types of Selenium Failures

Exception Type	What It Means
NoSuchElementFound	Element cannot be found on the page
StaleElementException	Element was found but is no longer on page (page refreshed)
TimeoutException	Element not found within the wait time
ElementNotInteractable	Element exists but cannot be clicked or interacted with
SessionNotCreated	Browser session cannot be created (driver issue)

10. What Selenium CANNOT Do

- Cannot test desktop applications
- Cannot do API testing
- Cannot test database
- Cannot generate test reports (use TestNG)
- Cannot handle all pop-ups and windows

💡 **For API Testing: Use POSTMAN Tool**

11. Automation vs Manual Testing

Aspect	Automation Testing	Manual Testing
Cost	Higher initial cost	Lower initial cost
Time	Faster (once setup)	Slower
Accuracy	Very accurate	Prone to human error
Reusability	Can run tests multiple times	Cannot reuse
When to Use	Regression testing	New features testing
Skills Needed	Coding skills needed	Domain knowledge

12. Types of Testing Frameworks

Frameworks are templates that help organize test code.

1. LINEAR FRAMEWORK (Record & Playback):

- Selenium IDE records user actions
- Plays back recorded actions as test

2. MODULAR FRAMEWORK:

- Divides application into modules
- Each module has its own test functions

3. DATA-DRIVEN FRAMEWORK:

- Tests run multiple times with different data
- Data comes from Excel, CSV, or database

4. KEYWORD-DRIVEN FRAMEWORK:

- Uses keywords (like Click, Type, Submit)
- Non-technical people can write tests

5. HYBRID FRAMEWORK:

- Combination of multiple frameworks
- Most powerful and flexible

PART 2: TestNG (NEXT GENERATION)

TestNG is a testing framework for Java that makes writing and running tests easier.

1. TestNG Annotations - Test Execution Flow

Annotations are special markers that tell Java when to run a method. Here's the execution order:

5. **@BeforeSuite**
   Runs ONCE before all tests in the suite
6. **@BeforeTest**
   Runs before each test tag
7. **@BeforeClass**
   Runs ONCE before all methods in the class
8. **@BeforeMethod**
   Runs before each test method
9. **@Test**
   The actual test method
10. **@AfterMethod**
    Runs after each test method
11. **@AfterClass**
    Runs ONCE after all methods in the class
12. **@AfterTest**
    Runs after each test tag
13. **@AfterSuite**

    Runs ONCE after all tests in the suite

2. Assertions in TestNG

Assertions are used to verify that tests are passing. There are TWO types:

🔴 HARD ASSERTIONS:

Hard assertions stop the test immediately if they fail. Here are all hard assertion types:

- assertEquals(actual, expected) - Check if two values are equal
- assertNotEquals(actual, expected) - Check if two values are NOT equal
- assertTrue(condition) - Check if condition is true
- assertFalse(condition) - Check if condition is false
- assertNull(value) - Check if value is null
- assertNotNull(value) - Check if value is NOT null
- assertSame(actual, expected) - Check if both objects are same (same memory reference)
- assertNotSame(actual, expected) - Check if objects are NOT same

🟢 SOFT ASSERTIONS:

Soft assertions do NOT stop the test if they fail. Test continues and reports all failures at the end.

- Use SAME assertions as hard assertions (above)
- Create SoftAssert object: SoftAssert softAssert = new SoftAssert();
- Use assertAll() at the end to get report of all failures

3. Difference Between Hard and Soft Assertions

Aspect	Hard Assertion	Soft Assertion
Stops Test?	YES - stops immediately	NO - continues
What Fails	One assertion	Multiple assertions
Reports	First failure only	All failures at end
When to Use	Critical checks	Non-critical checks
Example	Login page load check	Form field validations
assertAll() needed?	NO	YES - at the end

EXAMPLE - Hard Assertion (Stops on Failure):

Assert.assertEquals(actual, expected); // If fails, test STOPS here

EXAMPLE - Soft Assertion (Continues):

SoftAssert softAssert = new SoftAssert();

softAssert.assertEquals(actual1, expected1); // Fails but continues

softAssert.assertEquals(actual2, expected2); // Also checked

softAssert.assertAll(); // Reports ALL failures at end

GOOD LUCK WITH YOUR EXAM! 🎓

Remember: Practice > Theory